

SIMATS ENGINEERING

Assignment

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN

COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO1: Analyse the fundamentals of Object-Oriented Systems Development, including object basics, Object-Oriented Systems Development Life Cycle, Object-Oriented Methodologies, Model-Driven Development (MDD), and Agile practices.

CO2: Apply UML techniques including Use Case, Class, Interaction, State Transition, Activity, Package, Component, and Deployment Diagrams to model a software system.

CO3: Analyse objects, classes, attributes, methods, and relationships through Use Case Modelling, Object Analysis, Classification, and identification of object relationships.

Assignment Weightage: 100%

QUESTIONS: 1

Total Marks:100

Annexure A

Assignment

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Course Code & Name

CSA11 – Object Oriented Analysis and Design

Assignment Title	Object-Oriented Analysis and UML-Based Design of an Intelligent Food Waste Reduction and Redistribution System (IFWRRS)
Course Outcome(s) – CO	CO1: Analyse Object-Oriented Systems Development concepts and methodologies to understand and model a real-world software problem. CO2: Apply UML modelling techniques including Use Case, Class, Interaction, State Transition, Activity, Package, Component, and Deployment Diagrams. CO3: Analyse objects, classes, attributes, methods, and relationships through Use Case Modelling, Object Analysis, Classification, and relationship identification.
Bloom's Taxonomy Level	BL3 – Apply, BL4 – Analyse, BL5 – Evaluate, BL6 – Create
SDG Mapping (SDG 1–17)	SDG 2 – Zero Hunger SDG 11 – Sustainable Cities and Communities SDG 12 – Responsible Consumption and Production SDG 13 – Climate Action

Problem Statement

Analyse and develop an Object-Oriented Analysis and Design model for an Intelligent Food Waste Reduction and Redistribution System (IFWRRS) that connects restaurants, hotels, supermarkets, food suppliers, charitable organizations, delivery partners, and administrators to reduce avoidable food waste and facilitate timely redistribution of surplus food.

The system shall allow registered food providers to record surplus food, specify quantity, type, preparation time, expiry period, storage requirements, and pickup location. The system shall identify suitable recipient organizations based on food requirements, location, quantity, urgency, and availability.

Recipient organizations shall be able to view available food donations, request suitable items, confirm collection, and update distribution status. Delivery partners shall receive pickup and delivery assignments and update the transportation status. Administrators shall monitor food donations, organizations, deliveries, users, and system reports.

The system shall generate notifications when surplus food is approaching its expiry period and support intelligent matching between food providers and recipient organizations. Working from this scenario, students shall identify actors, objects, classes, attributes, methods, and relationships and develop Use Case, Class, Sequence, Activity, State Transition, Package, Component, and Deployment Diagrams.

Students shall analyse object responsibilities and apply suitable design principles and design patterns to achieve high cohesion, low coupling, reusability, maintainability, and scalability.

Constraints

The design shall clearly distinguish different stakeholders and their responsibilities. All identified classes, attributes, operations, and relationships shall be traceable to the identified use cases.

The system shall consider food expiry, notification timing, matching criteria, delivery status, user roles, data consistency, and transaction management. UML diagrams shall be mutually consistent and shall accurately represent the proposed system behaviour.

The final submission shall include assumptions, constraints, object responsibility analysis, design-pattern justification, and traceability between requirements, use cases, and design elements.

Tools Can Be Used

- StarUML
- Visual Paradigm
- Draw.io
- PlantUML
- Papyrus Tools
- ArgoUML
- Enterprise Architect
- Lucidchart

Assessment Rubrics

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Problem Understanding, Object Identification & Requirements Analysis (CO1)	10	Clearly analyses the problem domain and accurately identifies all major actors, objects, system responsibilities, and requirements from the given scenario.	Identifies most actors, objects, and requirements with minor gaps or inaccuracies.	Identifies basic actors and objects but analysis is incomplete or contains some inconsistencies.	Poor understanding of the problem; major actors, objects, and requirements are missing or incorrectly identified.
Use Case Modelling & Actor Identification (CO2)	10	Correctly identifies all relevant actors and use cases; Use Case Diagram clearly represents system boundary, relationships, and interactions.	Most actors and use cases are correctly identified with minor modelling errors.	Basic actors and use cases are identified, but relationships or system boundary need improvement.	Actors and use cases are poorly identified; diagram is incomplete or inconsistent.
Class Identification, Attributes, Methods & Relationships (CO2/CO3)	15	Classes, attributes, methods, associations, aggregation/composition, inheritance, and dependencies are	Class structure is mostly correct with minor errors in attributes,	Basic classes and relationships are identified but lack detail or contain several	Classes, attributes, methods, or relationships are missing, inappropriate, or

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
		accurately identified and logically justified.	methods, or relationships.	modelling issues.	poorly structured.
UML Behavioural & Interaction Modelling (CO2)	15	Sequence, Activity, and State Transition Diagrams accurately represent system behaviour and are fully consistent with the identified use cases and classes.	Behavioural diagrams are mostly correct with minor inconsistencies.	Basic behavioural diagrams are provided but some interactions, activities, or states are incomplete.	Behavioural diagrams are missing, unclear, or inconsistent with the proposed system.
UML Architectural Modelling & Design Evaluation (CO3)	15	Package, Component, and Deployment Diagrams clearly represent modular structure, system components, dependencies, and deployment environment with strong justification.	Architectural diagrams are mostly correct with minor omissions or modelling errors.	Basic architectural representation is provided but lacks clarity or sufficient justification.	Architectural diagrams are incomplete, inappropriate, or inconsistent with the system design.
Object Responsibilities, Design Principles, Axioms & Patterns (CO3)	15	Responsibilities are appropriately assigned; design axioms, cohesion, coupling, encapsulation, abstraction, and suitable design patterns are clearly justified and effectively applied.	Design principles and patterns are appropriately applied with minor limitations in justification.	Some design principles are identified, but application or justification is limited.	Design principles, axioms, responsibilities, or patterns are absent or incorrectly applied.
Consistency, Traceability & Overall Design Quality (CO3)	10	All UML models are mutually consistent, requirements are traceable to use cases/classes, and the overall design is complete, feasible, scalable, and maintainable.	Design is largely consistent and traceable with minor gaps.	Some consistency and traceability are demonstrated, but important links are missing.	Major inconsistencies exist between requirements, use cases, classes, and UML diagrams.
Reflection, SDG Relevance & Learning Outcomes	10	Provides insightful reflection on design decisions, challenges, OOAD concepts learned, and clearly establishes the relevance of the selected SDGs.	Provides good reflection with reasonable connection to design decisions and SDGs.	Reflection is present but generic with limited discussion of learning and SDG relevance.	Reflection is missing, superficial, or unrelated to the assignment and SDGs.

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Total	100				

Deliverables

To receive full credit, each submission must include the following:

1. **Problem Analysis and Requirement Summary**
2. **Actor Identification**
3. **Use Case Descriptions**
4. **Use Case Diagram**
5. **Object and Class Identification**
6. **Class Diagram**
7. **Sequence/Interaction Diagram(s)**
8. **Activity Diagram**
9. **State Transition Diagram**
10. **Package Diagram**
11. **Component Diagram**
12. **Deployment Diagram**
13. **Object Responsibility Analysis**
14. **Design Axioms/Principles and Design Pattern Justification**
15. **Assumptions and Constraints**
16. **Requirement-to-Use Case/Class Traceability**
17. **Implementation/Prototype and Results**
18. **Reflection on Design Decisions and Learning Outcomes**
19. **GitHub Upload**